



FY PROJECT — REQUIREMENTS DOCUMENT

Integrated Alumni Analytics and Employability Prediction System

Using Machine Learning and Deep Learning Techniques



Document Type: Complete Data & Project Requirements

Project Scope: BS, MS, PhD — Business Analytics & Project Management

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1. Data Requirements

Your project requires **three major categories** of data. Below is a detailed breakdown of every field needed to build the alumni database, train ML/DL models, and power the analytics dashboard.

1.1 Student Academic Data (Core Database)

This is the foundation of your system. You need records for all **BS, MS, PhD** students in **Business Analytics** and **Project Management**.

FIELD	TYPE	DESCRIPTION	EXAMPLE
student_id	String (PK)	Unique identifier	BA-BS-2020-001
full_name	String	Student's full name	Ahmed Khan
email	String	Contact email	ahmed@email.com
phone	String	Contact number	+92-300-XXXXXX
profile_photo_url	String	LinkedIn / Profile photo	URL
program	Enum	Business Analytics / Project Management	Business Analytics
degree_level	Enum	BS / MS / PhD	BS
enrollment_year	Integer	Year of admission	2020
graduation_year	Integer	Year of graduation	2024
graduation_status	Enum	Graduated / Enrolled / Dropped	Graduated
cgpa	Float	Cumulative GPA (0-4.0)	3.45
semester_wise_gpa	JSON/Array	GPA for each semester	[3.2, 3.5, 3.6, ...]
thesis_title	String	For MS/PhD students	"ML in Supply Chain"
thesis_supervisor	String	Supervisor name	Dr. Ali
total_credit_hours	Integer	Credits completed	132
batch	String	Batch identifier	Fall 2020

1.2 Skills & Competency Data

This is **critical for ML/DL prediction**. You need to capture what skills each student acquired during their studies.

FIELD	TYPE	DESCRIPTION	EXAMPLE
technical_skills	Array	Programming, tools, platforms	["Python", "SQL", "Tableau", "Power BI"]
soft_skills	Array	Communication, leadership, etc.	["Leadership", "Team Management"]
certifications	Array of Objects	Professional certifications	[{"name": "PMP", "issuer": "PMI", "year": 2023}]
courses_taken	Array of Objects	Key courses with grades	[{"course": "Data Mining", "grade": "A"}]
specialization_area	String	Focus area within program	"Supply Chain Analytics"
internship_experience	Array of Objects	Internships during studies	[{"company": "ABC", "duration": "3 months", "role": "Data Analyst Intern"}]
projects_completed	Array of Objects	Academic / industry projects	[{"title": "Sales Forecasting", "type": "Industry"}]
research_publications	Integer	Number of papers published	2
extracurricular_activities	Array	Societies, competitions, etc.	["IEEE Member", "Hackathon Winner"]
languages_known	Array	Programming & spoken languages	["Python", "R", "English", "Urdu"]
tools_proficiency	JSON	Tool proficiency levels (1-5)	{"Excel": 5, "Python": 4, "SPSS": 3}

1.3 Employment & Career Data (Post-Graduation)

This data is needed for the **monitoring dashboard** and serves as **labels for your ML models**.

FIELD	TYPE	DESCRIPTION	EXAMPLE
employment_status	Enum	Employed / Unemployed / Self-Employed / Further Studies	Employed
current_company	String	Company name	Systems Limited
company_type	Enum	MNC / Startup / Government / NGO / Academic	MNC
company_industry	String	Industry sector	IT & Consulting
job_title	String	Current designation	Data Analyst
job_level	Enum	Entry / Mid / Senior / Executive	Entry

FIELD	TYPE	DESCRIPTION	EXAMPLE
monthly_salary_range	Enum	Salary bracket	PKR 80K-120K
employment_start_date	Date	When they started working	2024-07-01
time_to_first_job	Integer	Months after graduation to first job	2
job_location	String	City, Country	Lahore, Pakistan
is_job_relevant	Boolean	Is the job related to their degree?	true
linkedin_url	String	LinkedIn profile link	URL
career_history	Array of Objects	Previous positions	[{"company": "XYZ", "role": "Intern", "duration": "6 months"}]
employer_satisfaction	Integer (1-5)	Employer feedback if available	4
alumni_feedback	Text	Self-reported career satisfaction	"Very satisfied..."

2. Data Collection Methods

 IMPORTANT — MINIMUM DATA REQUIREMENTS

You need at least **200-500+ records** for meaningful ML predictions. For deep learning, aim for **1000+ records** or use data augmentation techniques (SMOTE, GANs).

METHOD	DESCRIPTION	PRIORITY
University Records	Academic data from the registrar / admin office	● Must Have
Alumni Surveys	Google Forms / Microsoft Forms for career data	● Must Have
LinkedIn Scraping	Extract current positions (with consent)	● Nice to Have
Department Records	Course catalogs, skill mappings per course	● Must Have
Employer Feedback	Partner companies hiring your alumni	● Optional
Synthetic Data	SMOTE / GAN-generated for model training	● Nice to Have

3. ML/DL Model Requirements

3.1 Prediction Tasks

TASK	TYPE	INPUT FEATURES	OUTPUT
Employability Score	Regression	Skills, CGPA, Projects, Certs	Score (0-100%)
Job Placement Prediction	Classification	Academic + Skill data	Employed / Unemployed
Salary Range Prediction	Multi-class Classification	All features	Salary bracket
Time to Employment	Regression	Skills, CGPA, Internships	Months
Industry Sector Prediction	Multi-class Classification	Specialization, Skills	Industry type
Job Role Recommendation	Recommendation	Skill profile	Top-5 roles

3.2 Recommended Models

Machine Learning Models

MODEL	USE CASE	WHY
Random Forest	Employability Classification	Handles mixed features, interpretable
XGBoost / LightGBM	Salary & Employability Prediction	High accuracy on tabular data
Logistic Regression	Baseline binary classification	Simple, interpretable baseline
SVM	Small dataset classification	Works well with limited data
K-Nearest Neighbors	Job role matching	Similarity-based recommendations

Deep Learning Models

MODEL	USE CASE	WHY
Feed-Forward Neural Network (ANN)	Employability score prediction	Captures non-linear patterns
LSTM / GRU	Career trajectory prediction	Sequential semester data analysis
Autoencoders	Skill gap analysis	Dimensionality reduction & anomaly detection
Transformer-based (BERT)	Resume / feedback text analysis	NLP on alumni feedback text

3.3 Feature Engineering Requirements

These are the key derived features that should be engineered from raw student data before feeding into models:

Key Derived Features:

skill_diversity_score

→

Count of unique skill categories

└─

academic_consistency

→

Variance of semester GPAs

└─

practical_exposure

→

(Internships + Projects) weighted score

└─

certification_value

→

Weighted sum of certifications

└─

research_intensity

→

Publications / Years of study

└─

program_completion_rate

→

Credits completed / Required credits

└─

network_strength

→

Extracurricular + Online presence score

└─

industry_alignment

→

Skill match with industry demands

4. Technical Stack Requirements

4.1 Software Requirements

Component	Technology	Purpose
Frontend	React.js / Next.js	Dashboard UI
Backend	Python (Flask / FastAPI) or Node.js	API & Business Logic
Database	PostgreSQL / MongoDB	Student & Alumni data storage
ML/DL Framework	scikit-learn, TensorFlow/Keras, PyTorch	Model training
Data Processing	Pandas, NumPy	Data cleaning & transformation
Visualization	Chart.js, D3.js, Plotly	Dashboard charts
Deployment	Docker + AWS / Heroku / Vercel	Hosting
Version Control	Git + GitHub	Code management

4.2 Hardware Requirements

Component	Minimum	Recommended
RAM	8 GB	16 GB
Storage	50 GB SSD	100 GB SSD
GPU	Not required (for ML)	NVIDIA GPU (for DL training)
CPU	Intel i5 / Ryzen 5	Intel i7 / Ryzen 7

Component	Minimum	Recommended
Internet	Required	Stable broadband



TIP

Use **Google Colab Pro** or **Kaggle Notebooks** for free GPU access during DL model training if you don't have a dedicated GPU.

5. Project Architecture

The system follows a layered architecture with five distinct stages from data collection to dashboard presentation.



6. Dashboard Modules Required

MODULE	DESCRIPTION	KEY VISUALIZATIONS
Overview Dashboard	High-level KPIs and stats	Cards, Gauges, Summary charts
Alumni Directory	Searchable list of all alumni	Table with filters & search
Employability Predictor	Input student profile → Get prediction	Form + Score gauge + Recommendations

Module	Description	Key Visualizations
Employment Analytics	Where alumni are working	Bar charts, Pie charts, Maps
Skill Analytics	Most in-demand skills	Word clouds, Heatmaps, Radar charts
Program Comparison	BS vs MS vs PhD outcomes	Comparative bar charts
Geographic Distribution	Where alumni are located	Interactive map
Trend Analysis	Year-over-year employment trends	Line charts, Area charts
Admin Panel	Add / Edit / Delete alumni records	CRUD interface

7. Data Privacy & Compliance

 WARNING — PERSONAL DATA HANDLING

Since you're handling personal data of students and alumni, you must ensure compliance with data protection standards:

- Written consent from alumni for data collection
- Data anonymization for model training
- Secure storage with encryption
- Role-based access control on the dashboard
- Compliance with university data policies

8. Minimum Viable Product (MVP) Checklist

Complete these tasks to deliver a functional MVP of the system:

- ☐ Collect academic data for at least 200 students across all programs
- ☐ Design and distribute alumni survey form
- ☐ Clean and preprocess collected data
- ☐ Perform exploratory data analysis (EDA)
- ☐ Engineer features from raw data
- ☐ Train and compare at least 3 ML models

- ☐ Train at least 1 DL model (ANN / LSTM)
- ☐ Build REST API for model predictions
- ☐ Design and develop the web dashboard
- ☐ Integrate prediction API with dashboard
- ☐ Test with real alumni data
- ☐ Document the entire system

9. Suggested Timeline

A proposed 16-week development schedule for the entire project:

PHASE	DURATION	ACTIVITIES
Phase 1: Research & Data Collection	Weeks 1–3	Literature review, survey design, data gathering
Phase 2: Data Processing	Weeks 4–5	Cleaning, EDA, feature engineering
Phase 3: Model Development	Weeks 6–9	ML/DL model training, evaluation, tuning
Phase 4: Web Development	Weeks 10–13	Dashboard design, API development
Phase 5: Integration & Testing	Weeks 14–15	End-to-end testing, bug fixes
Phase 6: Documentation & Defense	Week 16	Report writing, presentation preparation